0309 HW 113705053

10.3

17 a. 假設 σ^2 相等 ⇒ F 檢定

$$F = \frac{S_{大}^2}{S_{小}^2} = \frac{2.78095}{0.17143} = 15.97054 \text{ (約大?)}$$

$$F_{14,14,0.05/2} = 2.9786$$

$\because 15.97054 > 2.9786$

$\therefore$ Reject $H_0: \sigma_1^2 = \sigma_2^2$

⇒ Not willing to assume #

b. $H_0: \mu_1 - \mu_2 = 0$ $H_a: \mu_1 - \mu_2 \neq 0$

unequal variance

$$T = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}}} = \frac{3.73 - 4.80}{\sqrt{\frac{2.78095}{15} + \frac{0.17143}{15}}}$$

$$= -2.4118$$

$$df = \left\lfloor \frac{\left(\frac{S_1^2}{n_1} + \frac{S_2^2}{n_2}\right)^2}{\frac{(S_1^2/n_1)^2}{n_1 - 1} + \frac{(S_2^2/n_2)^2}{n_2 - 1}} \right\rfloor = \left\lfloor \frac{\left(\frac{2.78095}{15} + \frac{0.17143}{15}\right)^2}{\frac{\left(\frac{2.78095}{15}\right)^2}{14} + \frac{\left(\frac{0.17143}{15}\right)^2}{14}} \right\rfloor$$

$$= 15 \text{ \#}$$

$$t_{15,0.025} = 2.1314$$

$|-2.4118| > 2.1314$ ⇒ reject H_0

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10.3

20. $\bar{\mu}_A = 5.0 \quad S_A^2 = 0.015$

$\bar{\mu}_B = 4.86 \quad S_B^2 = 0.013$

a. $H_0: \mu_A = \mu_B \quad , \quad \alpha = 0.05$

$H_a: \mu_A > \mu_B$

$$F = \frac{S_1^2}{S_2^2} = \frac{0.015}{0.013} = 1.923$$

$F_{4,4,0.025} = 9.6045$

$\because 1.923 < 9.6045$

$\therefore$ Non-reject H_0

$\Rightarrow$ No sufficient evidence indicate...

$\Rightarrow$ 可以假設 variance 相等.

b. $\bar{x}_1 - \bar{x}_2 \pm t_{n_1+n_2-2,\, \alpha/2} \cdot \sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$

95% C.I. $= 5 - 4.86 \pm t_{8,0.025} \cdot \sqrt{S_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}$

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$$

$$= \frac{4 \cdot 0.015 + 4 \times 0.013}{8}$$

$$= 0.019$$

$= 0.14 \pm 2.3060 \sqrt{0.019 \times \frac{1}{5}}$

$= [-0.061, 0.3410]$

$\Rightarrow$ non-Reject H_0

10.4

9.

	1	2	3	4	5
d_{i-2}	0.1	0.1	0	0.2	-0.1

$n=5$

$\bar{d} = 0.06$

$s_d^2 = 0.013$

$$T = \frac{\bar{d}-0}{s_d/\sqrt{n}} = \frac{0.06}{\sqrt{0.013}/\sqrt{5}} = 1.1767$$

$H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 \neq 0$

$df: n-1 = 4$

$t_{4,0.025} = 2.776$

$\because 1.1767 < 2.776$

$\therefore$ Non-reject H_0

$$95\%\ C.I. = \bar{d} \pm t_{4,0.025} \cdot \frac{s_d}{\sqrt{n}}$$

$$= 0.06 \pm 2.776 \times \sqrt{\frac{0.013}{5}}$$

$$= [-0.815, 0.2015]$$ 包含 0

$\Rightarrow$ Non-reject H_0

10.4

20 d with - without

1	15	6	8	11	9	26	13
2	15	7	12	12	4	17	4
3	15	8	8	13	10	18	7
4	9	9	10	14	4	19	7
5	16	10	12	15	17	20	10

$\bar{d} = 10.25$

$s_d = 4.05$

$H_0: \mu_1 - \mu_2 = 0$

$H_a: \mu_1 - \mu_2 > 0$

$T = \frac{\bar{d}}{s/\sqrt{n}} = \frac{10.25}{4.05/\sqrt{20}} = 11.3184$

$T_{19, 0.025} = 2.0930$

$\because 11.3184 > 2.093$

$\therefore$ Reject H_0

$\Rightarrow$ Yes

10.4

22 d_{A-B} .06 -0.18 -0.14 -0.11 -0.04 -0.19

$\bar{d} = -0.1$

$S_d = 0.0953$

a. $T = \frac{\bar{d}}{S_d/\sqrt{n}} = \frac{-0.1}{0.0953/\sqrt{6}} = -2.5703$

$T_{5,0.025} = 2.5706$

$\because |-2.5703| < 2.5706$

$\therefore$ non-reject H_0

$\Rightarrow N_0$

b 95% C.I $= d \pm t_{4,0.025} \frac{S_d}{\sqrt{n}}$

$= -0.1 \pm 2.5706 \frac{0.0953}{\sqrt{6}}$

$= [-0.200 \quad , 0.000012]$

$\ni 0$

$\Rightarrow$ Non-reject H_0

10.5

12 a. $\mu_{com} = \cancel{21.1765}\ 22.5$

$S_{com} = 4.705$

$\mu_{yard} = 12.99$

$S_{yard} = 21.531$

b. 95% C.I. for σ^2_{com}

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\alpha/2}}\right]$$

$$= \left[\frac{15\cdot(4.705)^2}{\chi^2_{15,0.05/2}}, \frac{15(4.705)^2}{\chi^2_{15,1-0.05/2}}\right]$$

$$= \left[\frac{332.0554}{27.4884}, \frac{332.0554}{6.2621}\right]$$

$$= [12.08, 53.0262]$$

c. 95% C.I. for σ^2_{yard}

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1,\alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1,1-\alpha/2}}\right]$$

$$= \left[\frac{15\cdot(21.531)^2}{27.4884}, \frac{15\cdot(21.531)^2}{6.2621}\right]$$

$$= [3.496, 15.346]$$

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σ 的 95% C.I

$= [\sqrt{3.496}, \sqrt{15.346}]$

$= [1.87, 3.92]$

d $H_0: \sigma = 4$ $\quad \alpha = 0.05$

$H_a: \sigma \neq 4$

$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{15 \cdot (2.531)^2}{16} = 6.06$

$\chi^2_{15, 0.975} = 27.4884$

$\because 6.06 < 27.4884$

$\therefore$ Non-reject H_0.

10.5

13. $\mu = 352.5$

$s = 1.915$

$s^2 = 3.67$

a $H_0: \sigma = 0.7$

$H_a: \sigma > 0.7$, $\alpha = 0.05$

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{3 \cdot 3.67}{0.49} = 22.47$$

$\chi^2_{3, 0.05} = 7.8147$

$\because 22.47 > 7.8147$

$\therefore$ Reject H_0

b 90% C.I. for σ^2

$$\left[\frac{(n-1)s^2}{\chi^2_{n-1, \alpha/2}}, \frac{(n-1)s^2}{\chi^2_{n-1, 1-\alpha/2}}\right]$$

$\alpha/2 = 0.05$, $1-\alpha/2 = 0.95$

$$= \left[\frac{3 \cdot 3.67}{7.8147}, \frac{3 \cdot 3.67}{0.3518}\right]$$

$= [1.409, 31.296] > 0.49$

$\Rightarrow$ Reject H_0

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